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Ideals defined by matrices and a certain complex associated with them

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For each matrix, whose elements belong to a commutative ring with an identity element, there is defined a free complex. This complex is a generalization of the standard Koszul complex, which corresponds to the case of a matrix with only a single row. The applications are to certain ideals defined by the maximal subdeterminants of a matrix. It is found that such an ideal has finite projective dimension whenever its grade reaches a certain greatest value (depending on the dimensions of the matrix) and that, in these circumstances, the complex provides a free resolution of the correct length. For semi-regular (= Macaulay–Cohen) rings this leads to a theorem on unmixed ideals. In the case of arbitrary Noetherian rings, a general theorem on rank is proved.

1. Introduction

Let R be a commutative ring with an identity element and let $a_1, a_2, \ldots, a_r$ be elements of R. It is known† that the Koszul complex generated by $a_1, a_2, \ldots, a_r$ has an intimate connexion with the ideal $(a_1, a_2, \ldots, a_r)$. Suppose now that

$$A = \begin{Vmatrix} a_{11} & a_{12} & \ldots & a_{1r} \\ a_{21} & a_{22} & \ldots & a_{2r} \\ . & . & \ldots & . \\ a_{s1} & a_{s2} & \ldots & a_{sr} \end{Vmatrix},$$

where $a_{ij} \in R$, is a given matrix and assume that $s \leqslant r$. The subdeterminants of order s will generate an ideal which will be denoted by (A), so that, in the special case $s = 1$, (A) is simply the ideal $(a_{11}, a_{12}, \ldots, a_{1r})$. A study of the properties of (A) leads one to the expectation that it should be possible to define a free complex, R^A say, which would reduce to the Koszul complex when $s = 1$, and, when $s > 1$, would still enjoy many of its properties. The present paper describes such a complex.

To illustrate the applications, let E be an R-module. Then, under appropriate conditions, it is found that the homology complex $H(R^A \otimes_R E)$ characterizes the grade‡ $\operatorname{gr}\{(A); E\}$ of (A) on E and that, in particular, $\operatorname{gr}\{(A); E\}$ never exceeds $r-s+1$. In the special situation $E = R$, this leads to the following result: *If* (i) R *is Noetherian*, (ii) A *is proper and* (iii) $\operatorname{gr}\{(A); R\} = r-s+1$, *then* $R/(A)$ *has projective dimension* $r-s+1$ *and* R^A *provides a free resolution with precisely* $r-s+1$ *terms*. This generalizes the corresponding theory for the Koszul complex which will be found in the early pages of Auslander & Buchsbaum (1958).

In the case of a semi-regular (= Macaulay–Cohen) ring, this leads to a theorem (due originally to F. S. Macaulay) which states that certain ideals defined by matrices

† For an account of this theory see Auslander & Buchsbaum (1958).

‡ This is also known as the 'co-dimension of E in (A)'.

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are unmixed in regard to rank. For less special Noetherian rings, grade and rank need separate consideration. Accordingly, a final section is devoted to a general theorem on rank. This shows that *if the subdeterminants of order t (of the matrix A) generate a proper ideal I, then the rank of each minimal prime ideal of I is at most* $(r-t+1)(s-t+1)$.

2. The Koszul complex

We begin by recalling some facts about the Koszul complex. Unless otherwise stated, R denotes an arbitrary commutative ring with an identity element.

Let $a_1, a_2, \ldots, a_r$ be elements of R and let $X_1, X_2, \ldots, X_r$ be indeterminates. We denote by K the exterior R-algebra on $X_1, X_2, \ldots, X_r$, so that K is a graded algebra with components $K_0 = R$, $K_1, K_2, \ldots, K_r$, and, if $0 \leqslant n \leqslant r$, K_n has an R-base consisting of all products

$$X_{i_1} X_{i_2} \ldots X_{i_n} \quad (1 \leqslant i_1 < \ldots < i_n \leqslant r).$$

Next, using the elements $a_1, a_2, \ldots, a_r$, we turn K into a complex

$$\ldots \longrightarrow 0 \longrightarrow K_r \xrightarrow{\Delta} K_{r-1} \longrightarrow \ldots \longrightarrow K_1 \xrightarrow{\Delta} K_0 \longrightarrow 0 \longrightarrow \ldots$$

by putting $$\Delta(X_{i_1} X_{i_2} \ldots X_{i_n}) = \sum_{p=1}^{n} (-1)^{p+1} a_{i_p} X_{i_1} \ldots \hat{X}_{i_p} \ldots X_{i_n},$$

where the $\hat{}$ over X_{i_p} means that this term is to be omitted. Observe that, for $n = 1$, the mapping gives

$$\Delta(X_{i_1}) = a_{i_1}.$$

This complex is the Koszul complex generated by $a_1, \ldots, a_r$. In addition to being a complex, it is also a skew algebra and we have

$$\Delta(x_m y_n) = (\Delta x_m) y_n + (-1)^m x_m (\Delta y_n), \tag{2·1}$$

where x_m, y_n are arbitrary elements of K_m, K_n, respectively.

Now let i, j $(i \neq j)$ be integers between 1 and r, let u, λ belong to R and assume that u is a unit. Put

$$X'_k = \begin{cases} X_k & (k \neq i) \\ uX_i + \lambda X_j & (k = i) \end{cases} \tag{2·2}$$

and

$$a'_k = \begin{cases} a_k & (k \neq i) \\ ua_i + \lambda a_j & (k = i) \end{cases}.$$

Then the products $$X'_{j_1} X'_{j_2} \ldots X'_{j_n} \quad (1 \leqslant j_1 < \ldots < j_n \leqslant r)$$

form an R-base for K_n and

$$\Delta(X'_{j_1} X'_{j_2} \ldots X'_{j_n}) = \sum_{p=1}^{n} (-1)^{p+1} a'_{j_p} X'_{j_1} \ldots \hat{X}'_{j_p} \ldots X'_{j_n}.$$

This situation will be described by saying that *if* $a_1, a_2, \ldots, a_r$ *are replaced by* $a'_1, a'_2, \ldots, a'_r$, *then the Koszul complex is unchanged.* The same conclusion holds if we permute $a_1, a_2, \ldots, a_r$ in any way.

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3. The complex associated with a matrix

Consider a matrix

$$A = \begin{Vmatrix} a_{11} & a_{12} & \dots & a_{1r} \\ a_{21} & a_{22} & \dots & a_{2r} \\ \cdot & \cdot & \dots & \cdot \\ a_{s1} & a_{s2} & \dots & a_{sr} \end{Vmatrix} \quad (s \leqslant r), \tag{3·1}$$

where the elements a_{ij} belong to R. The subdeterminants of A of order s will generate an ideal. This ideal will be denoted by (A).

Let K be the exterior algebra generated by $X_1, X_2, \dots, X_r$, then the kth row of A determines a differentiation Δ_k on K; in fact

$$\Delta_k(X_{i_1} X_{i_2} \dots X_{i_n}) = \sum_{p=1}^{n} (-1)^{p+1} a_{ki_p} X_{i_1} \dots \hat{X}_{i_p} \dots X_{i_n}. \tag{3·2}$$

Furthermore, one easily verifies that

$$\Delta_k \Delta_h + \Delta_h \Delta_k = 0 \quad (h \neq k). \tag{3·3}$$

Next let $Y_1, Y_2, \dots Y_s$ be s new symbols and, in the polynomial ring $R[Y_1, Y_2, \dots, Y_s]$, denote by Φ_n the R-module consisting of all forms of degree n. We are now ready to describe the complex R^A. It has component modules $R_0^A, R_1^A, \dots, R_{r-s+1}^A$, where

$$\left.\begin{aligned} R_{q+1}^A &= K_{s+q} \otimes_R \Phi_q \quad (q = 0, 1, \dots, r-s), \\ R_0^A &= R, \end{aligned}\right\}$$

and therefore, as a complex, it has the form

$$\dots \longrightarrow 0 \longrightarrow R_{r-s+1}^A \xrightarrow{d} R_{r-s}^A \longrightarrow \dots \longrightarrow R_1^A \xrightarrow{d} R_0^A \longrightarrow 0 \longrightarrow \dots. \tag{3·4}$$

In order to explain how the differentiation homomorphism d works, we observe that if $q \geqslant 0$, then R_{q+1}^A has an R-base consisting of the elements

$$X_{i_1} \dots X_{i_{s+q}} \otimes Y_1^{\nu_1} \dots Y_s^{\nu_s} \quad (1 \leqslant i_1 < \dots < i_{s+q} \leqslant r;\ \nu_1 + \dots + \nu_s = q).$$

When $q > 0$ the result of operating with d on a member of such a basis is given by the formula

$$d(X_{i_1} \dots X_{i_{s+q}} \otimes Y_1^{\nu_1} \dots Y_s^{\nu_s}) = \Sigma_j^* \Delta_j(X_{i_1} \dots X_{i_{s+q}}) \otimes Y_1^{\nu_1} \dots Y_j^{\nu_j - 1} \dots Y_s^{\nu_s},$$

where the asterisk attached to the summation sign means that we sum only over those values of j for which $\nu_j > 0$. For $q = 0$ the formula

$$d(X_{i_1} \dots X_{i_s} \otimes 1) = \begin{Vmatrix} a_{1i_1} & a_{1i_2} & \dots & a_{1i_s} \\ a_{2i_1} & a_{2i_2} & \dots & a_{2i_s} \\ \cdot & \cdot & \dots & \cdot \\ a_{si_1} & a_{si_2} & \dots & a_{si_s} \end{Vmatrix}$$

is to be employed. With the use of (3·3), it is a straightforward matter to verify that $d^2 = 0$. We leave this verification to the reader.

R^A is a free complex and

$$d(R_1^A) = (A). \tag{3·5}$$

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When $s = 1$, that is when the matrix consists of a single row, R_q^A $(q \geqslant 1)$ has the products

$$X_{i_1} \dots X_{i_q} \otimes Y_1^{q-1} \quad (i_1 < i_2 < \dots < i_q)$$

as an R-base and the differentiation becomes

$$d(X_{i_1} \dots X_{i_q} \otimes Y_1^{q-1}) = \Delta_1(X_{i_1} \dots X_{i_q}) \otimes Y_1^{q-2}$$

for $q > 1$, while $d(X_{i_1} \otimes 1) = a_{1i_1}$. This shows that, when the matrix has only a single row, R^A is essentially the Koszul complex generated by that row.

Once again, let i, j $(i \neq j)$ be integers between 1 and r, let u, λ belong to R and assume that u is a unit. Denote by A' the matrix obtained by operating on the ith column of A in the following way: we first multiply it by u and then add λ times the jth column. By changing from $X_1, \dots, X_r$ to $X_1', \dots, X_r'$ (see (2·2)), we find that not only can $R_n^{A'}$ be identified with R_n^A $(n = 0, 1, \dots, r-s+1)$ but this can be done in such a way that the differentiations coincide. We express this by saying that *the operation on the i-th column of A (described above) does not change* R^A. *Note that* R^A *is also unchanged when we permute the columns of A in an arbitrary manner.*

Let us see how R^A behaves relative to the operation of forming fractions. To this end let S be a multiplicatively closed set of elements of R and assume that $0 \notin S$. The canonical mapping $R \to R_S$ transports each constituent a_{ij} of A into an element of R_S and thereby gives rise to a matrix A_S whose elements belong to the ring of fractions. Now both $(R^A)_S$ and $R_S^{A_S}$ are complexes over the ring R_S and a little reflection shows that they can be identified. We therefore write

$$(R^A)_S = (R_S)^{A_S}. \tag{3·6}$$

After these preliminaries, let E be an R-module and put

$$E^A = R^A \otimes E, \tag{3·7}$$

where, from now on, $\otimes$ without any subscript will denote a tensor product over R. E^A is a complex

$$\dots \to 0 \to E_{r-s+1}^A \to \dots \to E_1^A \to E_0^A \to 0 \to \dots$$

and, since R^A is free, each exact sequence

$$0 \to E' \to E \to E^* \to 0$$

of R-modules gives rise to an exact sequence

$$0 \to E'^A \to E^A \to E^{*A} \to 0$$

of complexes. Again, every operation on the columns of A which leaves R^A unchanged will also leave E^A unchanged.

Finally, we recall that if P is an R-module, then we have an R_S-isomorphism

$$P_S \otimes_{R_S} E_S \approx (P \otimes_R E)_S$$

in which (with a self-explanatory notation) $[p/c] \otimes [e/\gamma]$ corresponds to $[(p \otimes e)/c\gamma]$. Furthermore, this isomorphism is natural for R-homomorphisms of the type $P \to P'$. It therefore follows, from (3·6), that we can make the identification

$$(E^A)_S = (E_S)^{A_S} \tag{3·8}$$

this asserting that the two sides agree as *complexes* over the ring R_S.

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4. Basic properties of E^A

In this section, R denotes an arbitrary commutative ring with an identity element, E is an R-module and A is the $r \times s$ matrix described in (3·1).

Proposition 1. *If (A) annihilates a non-zero element of E, then $H_{r-s+1}(E^A) \neq 0$.*

Proof. Since $E^A_{r-s+2} = 0$, we have only to establish the existence of a non-zero cycle of degree $r-s+1$. Observe first that if $s = r$, then E^A is just

$$\ldots \to E \to E \to 0 \to \ldots,$$

where $E \to E$ consists of multiplication by the determinant of A. Thus, in this case, the assertion is obvious. From now on we assume that $s < r$.

Next suppose that the ideal $(a_{11}, \ldots, a_{1r})$ annihilates a non-zero element e' of E. Then $(X_1 \ldots X_r \otimes Y_1^{r-s}) \otimes e'$ is a non-zero cycle of degree $r-s+1$. We can therefore also exclude this special case.

Let A_t denote the matrix formed by the first t rows of A, and (A_t) the ideal generated by its subdeterminants of order t. We choose t to be as large as possible subject to the condition that, in E,

$$0\colon (A_t) = 0. \tag{4·1}$$

Then our hypotheses secure that $1 \leqslant t < s$. Also, there exists $e \in E$ such that $e \neq 0$ and $(A_{t+1})e = 0$.

Let D denote a subdeterminant, of order t, of the matrix A_t. We contend that *D can be chosen so that no power of it annihilates e.* For otherwise there would exist an integer n such that $(A_t)^n e = 0$ and this leads at once to a contradiction of (4·1). Without loss of generality, we can assume that

$$D = \begin{Vmatrix} a_{11} & a_{12} & \ldots & a_{1t} \\ a_{21} & a_{22} & \ldots & a_{2t} \\ \cdot & \cdot & \ldots & \cdot \\ a_{t1} & a_{t2} & \ldots & a_{tt} \end{Vmatrix} \tag{4·2}$$

has this property. Accordingly, no power of D belongs to the annihilator $\operatorname{Ann}(Re)$ of Re. In view of this, there exists a prime ideal P such that $\operatorname{Ann}(Re) \subseteq P$ and $D \notin P$. Now form fractions with respect to $S = R - P$.

Since, by (3·8),

$$(H_{r-s+1}(E^A))_S = H_{r-s+1}((E^A)_S) = H_{r-s+1}((E_S)^{A_S}),$$

it will suffice to show that $H_{r-s+1}((E_S)^{A_S}) \neq 0$. Now $(Re)_S \subseteq E_S$, $[e/1] \neq 0$ and

$$(A_S)_{t+1}[e/1] = (A_{t+1})_S[e/1] = 0.$$

Thus, by localizing at P, we find that, for the remainder of the proof, we can proceed under the following assumptions:

(i) R has only a single maximal ideal;

(ii) $e \neq 0$ and $(A_{t+1})e = 0$;

(iii) the determinant D, in (4·2), is a unit.

It will be convenient to apply a number of column transformations to the matrix A. We observe, in advance, that none of these will change either the complex E^A or the ideal (A_{t+1}). So far as D is concerned, it will remain a unit.

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Since D is a unit and R has only one maximal ideal, it follows that at least one of $a_{11}, a_{12}, \ldots, a_{1t}$ is a unit. Using allowable transformations, we bring the first row of A to the form $(1, 0, \ldots, 0)$. The fact that D is a unit now shows that one of $a_{22}, \ldots, a_{2t}$ is a unit hence, by further allowable column transformations, we arrange that the first two rows of A are brought to the form

$$\left\| \begin{matrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \end{matrix} \right\|.$$

Proceeding in this way, we transform A into

$$\left\| \begin{matrix} 1 & 0 & 0 & \ldots & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 & 0 & \ldots & 0 \\ . & . & . & \ldots & . & . & \ldots & . \\ . & . & . & \ldots & . & . & \ldots & . \\ 0 & 0 & . & \ldots & 1 & 0 & \ldots & 0 \\ a_{t+1,1} & a_{t+1,2} & . & \ldots & a_{t+1,t} & a_{t+1,t+1} & \ldots & a_{t+1,r} \\ . & . & . & \ldots & . & . & \ldots & . \\ a_{s,1} & a_{s,2} & . & \ldots & a_{s,t} & a_{s,t+1} & \ldots & a_{s,r} \end{matrix} \right\|$$

say, without changing E^A.

It is now a simple matter to write down a non-zero cycle of degree $r-s+1$. To simplify the notation, let $c_1, c_2, \ldots, c_r$ denote the elements of the $(t+1)$th row and put

$$\xi = \sum_{\nu_1+\ldots+\nu_{t+1}=r-s} (-1)^{\nu_1+\ldots+\nu_t} c_1^{\nu_1} \ldots c_t^{\nu_t} X_1 \ldots X_r \otimes Y_1^{\nu_1} \ldots Y_t^{\nu_t} Y_{t+1}^{\nu_{t+1}}.$$

Since the expression for ξ contains the term $X_1 \ldots X_r \otimes Y_{t+1}^{r-s}$, it follows that $\xi \otimes e \neq 0$. Furthermore, $(A_{t+1})e = 0$ and therefore $c_k e = 0$ for all $k > t$. Bearing this in mind one verifies, in a straightforward manner, that $d(\xi \otimes e) = 0$. This completes the proof.

R^A has an intimate connexion with similar complexes determined by certain submatrices of A. This connexion will now be explored as it provides the basis of an inductive proof of the next proposition as well as being of interest for its own sake. Accordingly, until we come to the statement of proposition 2, it will be assumed that $1 < s < r$. This is in order that the appropriate submatrices shall exist and have the right kind of shape.

Denote by L the matrix which results when the last column of A is deleted, and by M the matrix obtained when both the last column and the first row are removed. Thus

$$L = \left\| \begin{matrix} a_{11} & a_{12} & \ldots & a_{1,r-1} \\ a_{21} & a_{22} & \ldots & a_{2,r-1} \\ . & . & \ldots & . \\ a_{s1} & a_{s2} & \ldots & a_{s,r-1} \end{matrix} \right\| \tag{4·3}$$

and

$$M = \left\| \begin{matrix} a_{21} & a_{22} & \ldots & a_{2,r-1} \\ a_{31} & a_{32} & \ldots & a_{3,r-1} \\ . & . & \ldots & . \\ a_{s1} & a_{s2} & \ldots & a_{s,r-1} \end{matrix} \right\|. \tag{4·4}$$

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It is clear that R^L can be regarded as a subcomplex of R^A and, indeed, R^L_q is a direct summand of R^A_q. To describe the connexion with R^M, we introduce certain R-homomorphisms†

$$\mu_q\colon\ R^L_q \to R^M_q \quad (q = 1, 2, \ldots, r-s+1) \tag{4·5}$$

which will not be defined. However, since $R^L_{r-s+1} = 0$, it will suffice if we describe the mappings

$$\mu_{q+1}\colon\ R^L_{q+1} \to R^M_{q+1} \quad (q = 0, 1, \ldots, r-s-1).$$

To this end we recall that R^L_{q+1} has an R-base consisting of all products

$$X_{i_1} X_{i_2} \ldots X_{i_{s+q}} \otimes Y^{\nu}_1 Y^{\nu}_2 \ldots Y^{\nu}_s,$$

where $1 \leqslant i_1 < \ldots < i_{s+q} \leqslant r-1$ and $\nu_1 + \ldots + \nu_s = q$. It will be convenient to describe the corresponding base for R^M_{q+1} by

$$U_{j_1} U_{j_2} \ldots U_{j_{s+q-1}} \otimes V^{n_2}_2 V^{n_3}_3 \ldots V^{n_s}_s,$$

where now $1 \leqslant j_1 < j_2 < \ldots < j_{s+q-1} \leqslant r-1$ and $n_2 + \ldots + n_s = q$. With this notation, μ_{q+1} $(q = 0, 1, \ldots, r-s-1)$ is obtained by putting

$$\mu_{q+1}(X_{i_1} \ldots X_{i_{s+q}} \otimes Y^{\nu}_1 Y^{\nu}_2 \ldots Y^{\nu}_s) = \sum_{p=1}^{s+q} (-1)^{p+1} a_{1 i_p} U_{i_1} \ldots \hat{U}_{i_p} \ldots U_{i_{s+q}} \otimes V^{\nu}_1 V^{\nu}_2 \ldots V^{\nu}_s, \tag{4·6}$$

where it is to be understood that the right-hand side of (4·6) is zero if $\nu_1 \geqslant 1$. The special property of these mappings, to which we wish to draw attention, is that

$$d_{q+1}\mu_{q+1} + \mu_q d_{q+1} = 0 \quad (q = 1, 2, \ldots, r-s). \tag{4·7}$$

Once again, the verification consists of a straightforward computation which we leave to the reader.

To simplify the notation, we shall frequently omit the suffix when referring to a mapping μ_q. On this understanding, we construct a complex

$$\ldots \longrightarrow 0 \longrightarrow R^M_{r-s+1} \oplus R^L_{r-s} \xrightarrow{\delta} R^M_{r-s} \oplus R^L_{r-s-1} \longrightarrow \ldots \longrightarrow R^M_2 \oplus R^L_1 \xrightarrow{\delta} R^M_1 \longrightarrow 0 \ldots, \tag{4·8}$$

where for $2 < n \leqslant r-s+1$ we put

$$\delta(x^M_n, x^L_{n-1}) = (dx^M_n + \mu x^L_{n-1}, dx^L_{n-1}), \tag{4·9}$$

while for $n = 2$ the corresponding formula is

$$\delta(x^M_2, x^L_1) = dx^M_2 + \mu x^L_1. \tag{4·10}$$

Of course, it is to be understood that $x^M_n \in R^M_n$ and $x^L_n \in R^L_n$. It follows, from (4·7) that $\delta^2 = 0$ so that (4·8) really is a complex.

It is convenient to denote this complex by $R^{L,M}$, where the component modules $R^{L,M}_n$ are so numbered that $R^{L,M}_0 = 0$, $R^{L,M}_1 = R^M_1$ and $R^{L,M}_n = R^M_n \oplus R^L_{n-1}$ for all $n > 1$. Accordingly (4·8) becomes

$$\ldots \longrightarrow 0 \longrightarrow R^{L,M}_{r-s+1} \xrightarrow{\delta} R^{L,M}_{r-s} \longrightarrow \ldots \longrightarrow R^{L,M}_1 \longrightarrow 0 \ldots. \tag{4·11}$$

† Observe that there is no mapping μ_0.

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LEMMA 1. *It is possible to define a mapping* $R^A \to R^{L,M}$ *of complexes in such a way that*

$$0 \to R^L \to R^A \to R^{L,M} \to 0 \tag{4·12}$$

becomes an exact sequence of complexes. When this is done, each of the exact sequences

$$0 \to R_q^L \to R_q^A \to R_q^{L,M} \to 0$$

splits.

Remark. It is this lemma which describes the connexion between R^L, R^A and R^M, which was referred to earlier.

Proof. Observe that, since R_q^L is a direct summand of R_q^A, the final assertion presents no problem. Furthermore, because $R_0^{L,M} = 0$, it will suffice to define R-epimorphisms

$$\phi_{q+1}\colon\ R_{q+1}^A \to R_{q+1}^{L,M} \quad (q = 0, \ldots, r-s) \tag{4·13}$$

so that $\operatorname{Ker} \phi_{q+1} = R_{q+1}^L$ for $q = 0, \ldots, r-s$ and $\delta_{q+1}\phi_{q+1} = \phi_q \delta_{q+1}$ for $q = 1, \ldots, r-s$. We shall do this by describing the action of ϕ_{q+1} on the individual generators of R_{q+1}^A.

Assume that $0 \leqslant q \leqslant r-s$. Since ϕ_{q+1} is to vanish on R_{q+1}^L, it is only necessary to show how generators of the form

$$X_{i_1} \ldots X_{i_{s+q-1}} X_r \otimes Y_1^{\nu_1} \ldots Y_s^{\nu_s} \quad (1 \leqslant i_1 < \ldots < i_{s+q-1} \leqslant r-1;\ \nu_1 + \ldots + \nu_s = q)$$

are to be mapped. Now

$$\left.\begin{aligned} R_{q+1}^{L,M} &= R_{q+1}^M \oplus R_q^L \quad (0 < q \leqslant r-s) \\ &= R_1^M \qquad\qquad (q = 0) \end{aligned}\right\}$$

and therefore $R_{q+1}^{L,M}$ is a free module with a base consisting of all products

$$U_{j_1} \ldots U_{j_{s+q-1}} \otimes V_2^{n_2} \ldots V_s^{n_s} \quad (1 \leqslant j_1 < \ldots < j_{s+q-1} \leqslant r-1;\ n_2 + \ldots + n_s = q)$$

together with the products

$$X_{i_1} \ldots X_{i_{s+q-1}} \otimes Y_1^{\nu_1} \ldots Y_s^{\nu_s} \quad (1 \leqslant i_1 < \ldots < i_{s+q-1} \leqslant r-1;\ \nu_1 + \ldots + \nu_s = q-1)$$

it being understood that the second set is empty when $q = 0$. We now define ϕ_{q+1} by putting

$$\phi_{q+1}(X_{i_1} \ldots X_{i_{s+q-1}} X_r \otimes Y_1^{\nu_1} Y_2^{\nu_2} \ldots Y_s^{\nu_s}) = U_{i_1} \ldots U_{i_{s+q-1}} \otimes V_2^{\nu_2} \ldots V_s^{\nu_s}$$

if $\nu_1 = 0$, while if $\nu_1 > 0$ we put

$$\phi_{q+1}(X_{i_1} \ldots X_{i_{s+q-1}} X_r \otimes Y_1^{\nu_1} Y_2^{\nu_2} \ldots Y_s^{\nu_s}) = X_{i_1} \ldots X_{i_{s+q-1}} \otimes Y_1^{\nu_1 - 1} Y_2^{\nu_2} \ldots Y_s^{\nu_s}.$$

Straightforward computations show that the mappings $\phi_0, \phi_1, \ldots, \phi_{r-s+1}$ have all the required properties.

LEMMA 2. *Let* D *denote the determinant*

$$D = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1s} \\ a_{21} & a_{22} & \ldots & a_{2s} \\ \cdot & \cdot & \ldots & \cdot \\ a_{s1} & a_{s2} & \ldots & a_{ss} \end{vmatrix}. \tag{4·14}$$

Then $D^s R_1^{L,M} \subseteq \delta(R_2^{L,M})$.

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Proof. $R_1^{L,M} = R_1^M$ has a base consisting of all products $U_{i_1}U_{i_2}\dots U_{i_{s-1}}\otimes 1$, where $1 \leqslant i_1 < \dots < i_{s-1} \leqslant r-1$. We wish to show that, in every case,

$$D^s(U_{i_1}\dots U_{i_{s-1}}\otimes 1)$$

can be expressed in the form $dx_2^M + \mu x_1^L$ (see (4·10)). Denote by k the number of integers belonging to the set $i_1, \dots, i_{s-1}$ which do not occur among $1, 2, \dots, s$. We shall prove, by induction on k, that $D^{k+1}(U_{i_1}\dots U_{i_{s-1}}\otimes 1)$ can be expressed in the form $dx_2^M + \mu x_1^L$. This will clearly establish the lemma. Observe that

$$0 \leqslant k \leqslant \min(s-1, r-s+1).$$

To this end we use the relations

$$\mu(X_1X_2\dots X_s\otimes 1) = \sum_{p=1}^{s}(-1)^{p+1}a_{1p}(U_1\dots \hat{U}_p\dots U_s\otimes 1)$$

$$d(U_1U_2\dots U_s\otimes V_2) = \sum_{p=1}^{s}(-1)^{p+1}a_{2p}(U_1\dots \hat{U}_p\dots U_s\otimes 1)$$

$$\dots\dots\dots

belongs to $\mu(R_1^L)+d(R_2^M)$. If therefore we multiply through by D^k and apply the induction hypothesis, we obtain

$$D^{k+1}(U_{k+2}\dots U_s U_{j_1}\dots U_{j_k}\otimes 1)\in \mu(R_1^L)+d(R_2^M)$$

which completes the proof.

The next lemma is not new (see Auslander & Buchsbaum, proposition 1·5) but a short proof is included for the sake of completeness.

LEMMA 3. *Let $a_1, a_2, \dots, a_r$ belong to R and let K be the Koszul complex which they generate. If now E is an R-module, then $(a_1, \dots, a_r)\, H(K\otimes E)=0$.*

Proof. Let z_n be a cycle of $K\otimes E$ of degree n, then it can be expressed in the form

$$z_n = x_n^{(1)}\otimes e_1+\dots+x_n^{(t)}\otimes e_t \quad (x_n^{(j)}\in K_n;\ e_j\in E),$$

where $\Sigma_j dx_n^{(j)}\otimes e_j=0$. Accordingly

$$\begin{aligned} d(X_i x_n^{(1)}\otimes e_1+\dots+X_i x_n^{(t)}\otimes e_t) &= \Sigma_j d(X_i x_e^{(j)})\otimes e_j \\ &= \Sigma_j a_i x_n^{(j)}\otimes e_j-\Sigma_j X_i dx_n^{(j)}\otimes e_j \\ &= a_i z_n. \end{aligned}$$

It follows that $a_i H_n(K\otimes E)=0$ and, as this holds for $1\leqslant i\leqslant r$, the lemma follows.

After these preliminaries, let E be an R-module and put

$$E^{L,M}=R^{L,M}\otimes E \tag{4·15}$$

so that $E^{L,M}$ has a natural structure as a complex. Since, by Lemma 1, each exact sequence

$$0\to R_q^L\to R_q^A\to R_q^{L,M}\to 0$$

splits, this induces an exact sequence

$$0\to E^L\to E^A\to E^{L,M}\to 0$$

of complexes. This, in turn, gives rise to the exact homology sequence

$$\dots\to H_n(E^L)\to H_n(E^A)\to H_n(E^{L,M})\to H_{n-1}(E^L)\to\dots \tag{4·16}$$

which will be useful later. For the moment, we analyze the structure of $E^{L,M}$.

In the first place, the mapping (4·5) induces a homomorphism

$$E_q^L\to E_q^M \quad (q=1,2,\dots,r-s+1)$$

which we also denote by μ_q. This inherits the property described in (4·7) so that we have

$$d_{q+1}\mu_{q+1}+\mu_q d_{q+1}=0 \quad (q=1,2,\dots,r-s), \tag{4·17}$$

where now the d's refer to the differentiations of the complexes E^M and E^L. Next

$$E_n^{L,M}=(R_n^M\oplus R_{n-1}^L)\otimes E=E_n^M\oplus E_{n-1}^L \quad (n>1) \tag{4·18}$$

and

$$E_1^{L,M}=E_1^M, \quad E_0^{L,M}=0. \tag{4·19}$$

Moreover, if we use δ to denote the differentiation homomorphism on $E^{L,M}$, then (with a self-explanatory notation) (4·9) yields

$$\delta(e_n^M, e_{n-1}^L)=(de_n^M+\mu e_{n-1}^L, de_{n-1}^L) \tag{4·20}$$

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for $2 < n \leqslant r-s+1$, while (4·10) becomes

$$\delta(e_2^M, e_1^L) = de_2^M + \mu e_1^L. \tag{4·21}$$

Finally, if D denotes the determinant referred to in lemma 2, then, by that lemma,

$$D^s E_1^{L,M} \subseteq \delta(E_2^{L,M}). \tag{4·22}$$

We can now establish a second fundamental property of the complex E^A.

PROPOSITION 2. *There exists an integer $h = h(r,s)$ such that $(A)^h H(E^A) = 0$.*

Remark. This proposition states that a certain power of the ideal (A) annihilates all the homology modules of the complex E^A. Furthermore, the power to which (A) must be raised (for this purpose) is independent of the ring R, the module E and the individual elements of the matrix A. In other words, it is determined solely by r and s.

Proof. We observe first that if $s = 1$, then $(A)H(E^A) = 0$ for in this case the proposition reduces to lemma 3. Also, if $s = r$, then $(A)H(E^A) = 0$, because E^A is just the complex $\ldots \to 0 \to E \to E \to 0 \to \ldots$, where $E \to E$ consists of multiplication by the determinant of A.

We turn now to the general case where s, r $(s \leqslant r)$ are arbitrary, but we assume that the proposition has been proved for all pairs s', r' $(s' \leqslant r')$ for which either $s' < s$ or $s' = s$ and $r' < r$.

Consider the determinant

$$D = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1s} \\ a_{21} & a_{22} & \ldots & a_{2s} \\ \cdot & \cdot & \ldots & \cdot \\ a_{s1} & a_{s2} & \ldots & a_{ss} \end{vmatrix}.$$

We shall begin by showing that there exists an integer $m = m(r,s)$ such that $D^m H(E^A) = 0$. If $s = 1$ or $s = r$ there is no problem (take $m = 1$), so we assume that $1 < s < r$. Let L and M be the matrices (4·3) and (4·4), respectively. By hypothesis, we can find an integer $k = k(r,s)$ so that

$$(L)^k H(E^L) = 0 \quad \text{and} \quad (M)^k H(E^M) = 0.$$

But $D \in (L)$ and $D \in (M)$, consequently

$$D^k H(E^L) = 0 \quad \text{and} \quad D^k H(E^M) = 0.$$

Let us now turn our attention to $E^{L,M}$. Certainly $H_0(E^{L,M}) = 0$ and, by (4·22), $D^s H_1(E^{L,M}) = 0$. It will now be proved that

$$D^{2k} H_n(E^{L,M}) = 0 \quad (2 \leqslant n \leqslant r-s+1). \tag{4·23}$$

To this end let (e_n^M, e_{n-1}^L) be a cycle of $E^{L,M}$ of degree $n > 2$, then, by (4·20), $de_{n-1}^L = 0$ and $de_n^M + \mu e_{n-1}^L = 0$. Since $D^k H_{n-1}(E^L) = 0$, it follows that we can write $D^k e_{n-1}^L = de_n^L$, where $e_n^L \in E_n^L$. Accordingly,

$$\begin{aligned} 0 &= d(D^k e_n^M) + \mu(D^k e_{n-1}^L) \\ &= d(D^k e_n^M) + \mu de_n^L \\ &= d(D^k e_n^M - \mu e_n^L) \end{aligned}$$

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by virtue of (4·17). However, $D^k H_n(E^M) = 0$ consequently we can find $e^M_{n+1} \in E^M_{n+1}$ such that $D^k(D^k e^M_n - \mu e^L_n) = d e^M_{n+1}$, and therefore

$$D^{2k} e^M_n = d e^M_{n+1} + \mu(D^k e^L_n).$$

Thus
$$\begin{aligned}\delta(e^M_{n+1}, D^k e^L_n) &= (d e^M_{n+1} + \mu D^k e^L_n, d D^k e^L_n) \\ &= (D^{2k} e^M_n, D^{2k} e^L_{n-1}) \\ &= D^{2k}(e^M_n, e^L_{n-1})\end{aligned}$$

which proves (4·23) when $n > 2$. For $n = 2$ there is a minor complication. To deal with this, suppose that (e^M_2, e^L_1) is a cycle of degree 2. Then, by (4·21), $d e^M_2 + \mu e^L_1 = 0$. If we apply to this the mapping $d\colon E^M_1 \to E^M_0$, then we obtain $d\mu e^L_1 = 0$. However, E^M_0 and E^L_0 can both be identified with E and, if this done, then we find that $d\mu e^L_1 = d e^L_1$. Thus we also have $d e^L_1 = 0$ and now the argument proceeds as before.

Put $\nu = \max(2k, s)$, then ν depends only on r and s and $D^\nu H(E^{L,M}) = 0$. Furthermore, by (4·16), we have an exact sequence

$$H_n(E^L) \to H_n(E^A) \to H_n(E^{L,M}) \quad (n \text{ arbitrary})$$

and so, since $D^k H_n(E^L) = 0$ and $D^\nu H_n(E^{L,M}) = 0$, we have $D^{k+\nu} H_n(E^A) = 0$. Accordingly, if we put $m = k + \nu$, then $m = m(r, s)$ and $D^m H(E^A) = 0$.

Next, let D' be any subdeterminant of order s of the matrix A. By permuting the columns of A we can arrange that the columns of D' occupy the first s places. This operation will not change the complex E^A and therefore, by the result already obtained, $(D')^m H(E^A) = 0$.

Finally, the ideal (A) is generated by $\binom{r}{s}$ subdeterminants D'. Put $h = \binom{r}{s} m$, then $h = h(r, s)$ and $(A)^h H(E^A) = 0$.

Proposition 3. *Let E be an R-module and suppose that either* (i) $(A) = R$ or (ii) *E is finitely generated and $(A)E = E$. Then $H(E^A) = 0$.*

Proof. If condition (i) is satisfied, then the assertion follows at once from proposition 2. Now assume that $E = Re_1 + \ldots + Re_t$ and $(A)E = E$. Then $e_i = \Sigma_j \alpha_{ij} e_j$, where $\alpha_{ij} \in (A)$, and therefore the determinant $|\delta_{ij} - \alpha_{ij}|$ annihilates E. (Here $\delta_{ij} = 0$ if $i \neq j$ while $\delta_{ii} = 1$.) It follows that we can write $1 = \alpha + \beta$, where $\alpha \in (A)$ and $\beta E = 0$. However, by proposition 2, there exists an integer h such that $\alpha^h H(E^A) = 0$, consequently

$$H(E^A) = (\alpha + \beta)^h H(E^A) = \alpha^h H(E^A) = 0.$$

5. Applications

We shall now apply the results of the preceding sections to the theory of finitely generated modules over Noetherian rings.

Suppose then that R is Noetherian, let E be a finitely generated R-module and I an ideal such that $IE \neq E$. A sequence $u_1, u_2, \ldots, u_p$ of elements of R is said to be an *R-sequence on E contained in I* if (i) $u_1, u_2, \ldots, u_p$ are all in I and (ii) for each i $(1 \leqslant i \leqslant p)$, u_i is not a zero-divisor in $E/(u_1, \ldots, u_{i-1})E$. Such a sequence is said to be *maximal* if every element of I is a zero-divisor in $E/(u_1, \ldots, u_p)E$ which means,

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of course, that the sequence cannot be extended. It is known (Auslander & Buchsbaum 1958, p. 630) that every R-sequence on E contained in I can be extended to a (finite) maximal sequence of the same kind; moreover, any two maximal R-sequences (on E and contained in I) have the same number of terms. The number of terms in a maximal R-sequence is known as the *grade of I on E* or the *co-dimension of E in I*. In the present paper, this number will be denoted by $\operatorname{gr}(I; E)$.

THEOREM 1. *Let R be a Noetherian ring and E a finitely generated R-module. Further, let (A) be an ideal defined by an $s \times r$ matrix $(s \leqslant r)$ and assume that $A(E) \neq E$. If now q is the largest value of m such that $H_m(E^A) \neq 0$, then*

$$\operatorname{gr}\{(A); E\} + q = r - s + 1$$

and, in particular, $\operatorname{gr}\{(A); E\} \leqslant r - s + 1$.

Remark. The corresponding result for the Koszul complex is proved in Auslander & Buchsbaum (1958, theorem 1·7, p. 629). The argument given here is an adaptation of that proof.

Proof. Put $\operatorname{gr}\{(A); E\} = k$ and first consider the case $k = 0$. Then every element of (A) is a zero-divisor in E and therefore, since we are in the Noetherian case, there exists $e \in E$ such that $e \neq 0$ and $(A)e = 0$. Accordingly, by proposition 1, $q = r - s + 1$ which shows that the theorem is true in the present instance.

We now proceed by induction on k assuming that $k > 0$ and that the theorem has been proved for all smaller values of the inductive variable. Let $x_1, x_2, \ldots, x_k$ be a maximal R-sequence on E and contained in (A). Then

$$0 \to E \to E \to E/x_1E \to 0 \tag{5·1}$$

is an exact sequence, where $E \to E$ is the mapping produced by multiplication by x_1. Furthermore, R^A is a free complex, and so (5·1) induces an exact sequence

$$0 \to E^A \to E^A \to (E/x_1E)^A \to 0$$

of complexes. This, in turn, gives rise to the exact homology sequence

$$\ldots \to H_n(E^A) \to H_n(E^A) \to H_n((E/x_1E)^A) \to H_{n-1}(E^A) \to H_{n-1}(E^A) \to \ldots \tag{5·2}$$

in which each of the mappings $H_n(E^A) \to H_n(E^A)$ is also produced by multiplication by x_1.

Let q' be the largest value of m such that $H_m((E/x_1E)^A) \neq 0$. Since

$$\operatorname{gr}\{(A); E/x_1E\} = k - 1$$

the inductive hypothesis can be applied to show that

$$k - 1 + q' = r - s + 1. \tag{5·3}$$

However, if $n \geqslant q'$, then (5·2) shows that we have an exact sequence

$$0 \to H_n(E^A) \xrightarrow{x_1} H_n(E^A)$$

which means that x_1 is not a zero-divisor in $H_n(E^A)$. On the other hand, $x_1 \in (A)$ and therefore, by proposition 2, $x_1^h H_n(E^A) = 0$ for a suitable integer h. It follows that $H_n(E^A) = 0$ provided $n \geqslant q'$.

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Returning to (5·2), we see that

$$0 \to H_{q'}((E/x_1E)^A) \to H_{q'-1}(E^A)$$

is exact and, by the definition of q', the middle term is not zero. Thus $H_{q'-1}(E^A) \neq 0$ which shows that $q = q' - 1$. Accordingly, by (5·3), $k+q = r-s+1$ and the theorem is proved.

COROLLARY. *Let R be a Noetherian ring and E a finitely generated R-module for which $(A)E \neq E$. If now* $\operatorname{gr}\{(A); E\} = r-s+1$, *then*

$$0 \to E^A_{r-s+1} \to E^A_{r-s} \to \ldots \to E^A_1 \to E \to E/(A)E \to 0$$

is an exact sequence.

Proof. In the notation of the theorem, we have $q = 0$ consequently $H_n(E^A) = 0$ for all $n > 0$.

THEOREM 2. *Let R be a Noetherian ring and (A) a proper ideal defined by an $s \times r$ $(s \leqslant r)$ matrix. If now* $\operatorname{gr}\{(A); R\} = r-s+1$, *then*

$$0 \to R^A_{r-s+1} \to R^A_{r-s} \to \ldots \to R^A_1 \to R^A_0 \to R/(A) \to 0 \tag{5·4}$$

is a free resolution of $R/(A)$. Furthermore, under these conditions, $r-s+1$ is precisely the projective dimension of $R/(A)$.

Remark. A proof that, *when* $\operatorname{gr}\{(A); R\} = r-s+1$, *the projective dimension of $R/(A)$ is* $r-s+1$ is to be found in Eagon (1961). This proof did not exhibit an explicit free resolution of the module.

Proof. That (5·4) is exact follows from the corollary to theorem 1. This shows that the projective dimension of $R/(A)$ is at most $r-s+1$. Now let P be any prime ideal belonging to (A), then $\operatorname{gr}\{P; R\} \geqslant r-s+1$ and P belongs to the zero submodule of $R/(A)$. If now $dh_R R/(A)$ denotes the projective dimension of $R/(A)$, then, by a theorem due to Rees (1957, theorem 1·2, cor., p. 30), $\operatorname{gr}\{P; R\} \leqslant dh_R R/(A)$. This shows that

$$\operatorname{gr}\{P; R\} = dh_R R/(A) = r-s+1. \tag{5·5}$$

The additional information contained in (5·5) is sufficiently important to be stated formally in the following

COROLLARY. *Let R be Noetherian and (A) a proper ideal generated by the determinants of order s of an $s \times r$ $(s \leqslant r)$ matrix. If now* $\operatorname{gr}\{(A); R\} = r-s+1$, *then* $\operatorname{gr}\{P; R\} = r-s+1$ *for every prime ideal P belonging to (A).*

6. A THEOREM ON RANK

Suppose, for the moment, that R is a polynomial ring in m variables with coefficients in a field, and that A is an $s \times r$ $(s \leqslant r)$ matrix of elements of R. This particular ring has the property that $\operatorname{gr}(I; R) = \operatorname{rank} I$ for every proper ideal I. We can therfore apply theorem 1 and the corollary to theorem 2 to obtain the following results: *If (A) is a proper ideal, then rank $(A) \leqslant r-s+1$. Furthermore, if rank $(A) = r-s+1$, then all the prime ideals belonging to (A) also have rank equal to $r-s+1$.* This is a classical result due to Macaulay (1916, p. 54).

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More generally, the same theorem holds in any semi-regular (= Macaulay–Cohen) ring, because one has the same simple connexion between grade and rank. It may be noted that Macaulay's theorem makes no reference to homological concepts and, indeed, it can be proved (in the generalized form) without using homological arguments (see Northcott 1960).

Now let R be an *arbitrary* Noetherian ring and A an $s \times r$ ($s \leqslant r$) matrix composed of elements of R. One might conjecture that if (A) is a proper ideal, then all the minimal prime ideals of (A) have rank at most $r-s+1$. This is, in fact, true and is a special case of a more general theorem which will be proved in this section.

To this end, denote by I the ideal generated by all the subdeterminants of A of order t ($t \leqslant s$). It will be shown that, *if I is a proper ideal, then all its minimal prime ideals have rank at most* $(r-t+1)(s-t+1)$. This theorem is to be found in Eagon (1961).

We begin with a number of very simple observations. Let u and λ belong to R, suppose that u is a unit, and let C_i and C_j be two *distinct* columns of A. The operation of replacing C_i by $uC_i + \lambda C_j$ (while the remaining columns stay fixed) will be called an 'elementary column transformation'. We also refer to a permutation of the columns of A as an elementary column transformation. Correspondingly, we have two kinds of elementary row transformation. It is easily verified that *all elementary row and column transformations leave the ideal I unchanged.*

Suppose now that (i) I is proper and (ii) at least one element of A is a unit. Then $t > 1$ and, by means of elementary row and column transformations, A can be brought to the form

$$\begin{Vmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & B & \\ 0 & & & \end{Vmatrix}$$

without changing I. If we delete the first row and column of the new matrix, we are left with an $(s-1) \times (r-1)$ matrix B which has the property that its subdeterminants of order $t-1$ generate the original ideal I. This observation will be used, in an inductive argument, at a later stage.

Lemma 4. *Let R be a local ring with maximal ideal P and let J be a P-primary ideal. In the polynomial ring $R[x]$, let $J' \subseteq PR[x]$ be an ideal with the property that*

$$(J', x) = (JR[x], x).$$

Then J' has $PR[x]$ as one of its minimal prime ideals.

Proof. Since $PR[x]$ is a prime ideal, we can find a minimal prime ideal P' of J' such that $P' \subseteq PR[x]$. Next (P', x) is contained in the prime ideal $(PR[x], x)$, consequently there exists a minimal prime ideal P'' of (P', x) such that

$$(P', x) \subseteq P'' \subseteq (PR[x], x).$$

Now
$$JR[x] \subseteq (JR[x], x) = (J', x) \subseteq (P', x) \subseteq P'',$$

and therefore $J \subseteq P'' \cap R$ which shows that $P'' \cap R = P$. It follows that $(PR[x], x)$ is contained in P'' and this shows that, in fact, $P'' = (PR[x], x)$.

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Next, in the ring $R[x]/P'$, $(P',x)/P'$ is a principal ideal and P''/P' is one of its minimal prime ideals. This shows that P''/P', i.e. $(PR[x],x)/P'$, has rank at most equal to one. But $(PR[x],x) \supset PR[x] \supseteq P'$ is a chain of prime ideals, consequently $PR[x] = P'$ and the lemma is proved.

THEOREM 3. *Let R be Noetherian and suppose that the subdeterminants of order t, of the $s \times r$ ($s \leqslant r$) matrix A, generate a proper ideal I. If now P is any minimal prime ideal of I, then rank $P \leqslant (r-t+1)(s-t+1)$.*

Proof. We use induction on s. If $s = 1$, then also $t = 1$ and I is generated by r elements. It follows that rank $P \leqslant r$ which is the value of $(r-t+1)(s-t+1)$ in this special case.

We next assume that $s > 1$ and that the theorem has been proved for all smaller values of the inductive variable. By localizing at P, we find that we may assume (for the purposes of the proof) that R is a local ring with maximal ideal P and that I is P-primary.

If $t = 1$, then I is generated by rs elements consequently rank $P \leqslant rs$ which is the value of $(r-t+1)(s-t+1)$ for this case. Also, if one of the elements of A happens to be a unit, then I is generated by the subdeterminants, of order $t-1$, of an $(s-1) \times (r-1)$ matrix. In this case, the required result follows by the induction hypothesis. *We may therefore suppose that $t > 1$ and also that every element a_{ij} of A is contained in the maximal ideal P.*

Let x be an indeterminate, put

$$A' = \begin{Vmatrix} a_{11}+x & a_{12} & \dots & a_{1r} \\ a_{21} & a_{22} & \dots & a_{2r} \\ \cdot & \cdot & \dots & \cdot \\ a_{s1} & a_{s2} & \dots & a_{sr} \end{Vmatrix}$$

and denote by I' the ideal of $R[x]$ which is generated by the subdeterminants of order t of the matrix A'. Since $t > 1$ and $a_{ij} \in P$ for all i and j, we have $I' \subseteq PR[x]$. We also have $(I',x) = (IR[x],x)$ as may be seen by passing to the residue class ring $R[x]/(x)$. It therefore follows, from lemma 4, that $PR[x] = P'$ (say) is a minimal prime ideal of I'. Now rank P' = rank P and thus we see that we can consider A', I' and P' in place of A, I and P. This is an advantage because the element $a_{11}+x$ of A' does not belong to P'.

At this point we localize at P' thereby transforming A', I' and P' into A^*, I^* and P^* (say). Since rank P^* = rank P', it is enough to prove that rank

$$P^* \leqslant (r-t+1)(s-t+1).$$

However, one of the elements of A^* is a unit and we already know that, in such a case, the required result follows from the induction hypothesis. The proof is now complete.

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